

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024
FULL MARKS: 75

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) How do numerical approaches differ from analytical approaches to solving a problem?

4
(CO2)
(PO1)

Solution:

Any 2 valid differences will suffice. Some differences can be written as follows —

Criteria	Numerical Methods	Analytical Methods
Type of yielded solution	Approximate solution	Exact solution
Working Procedure	Keep iterating until the obtained solution is a “good enough” approximation.	Perform symbolic manipulations to arrive at an equation that yields the exact solution.
Example	Solving $ax^2 + bx + c = 0$ using the quadratic equation $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Solving $ax^2 + bx + c = 0$ using any root-finding algorithm e.g. the Bisection method.

Rubric:

- 2 points for each valid difference.

- b) Derive the formula for the Secant method geometrically.

6
(CO4)
(PO1)

Solution:

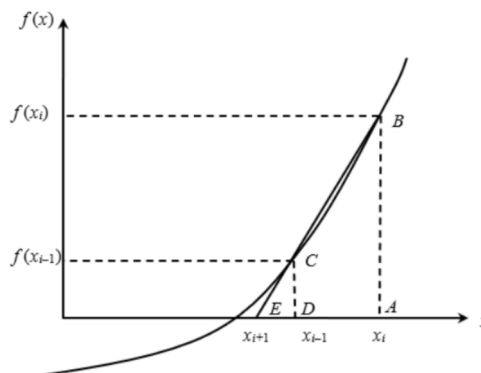


Figure 1: Geometrical representation of the Secant method.

Here, ABE and DCE are similar triangles.

Hence,

$$\begin{aligned}
 \frac{AB}{AE} &= \frac{DC}{DE} \\
 \frac{f(x_i)}{x_i - x_{i+1}} &= \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}} \\
 \Rightarrow f(x_i)(x_{i-1} - x_{i+1}) &= f(x_{i-1})(x_i - x_{i+1}) \\
 \Rightarrow f(x_i)x_{i-1} - x_{i+1}f(x_i) &= x_if(x_{i-1}) - x_{i+1}f(x_{i-1}) \\
 \Rightarrow x_{i+1}f(x_{i-1}) - x_{i+1}f(x_i) &= f(x_{i-1})x_i - f(x_i)x_{i-1} \\
 \Rightarrow x_{i+1} &= \frac{x_if(x_{i-1}) - f(x_i)x_{i-1}}{f(x_{i-1}) - f(x_i)} + x_i - x_i \\
 \Rightarrow x_{i+1} &= x_i + \frac{x_if(x_{i-1}) - f(x_i)x_{i-1} - x_if(x_{i-1}) + x_if(x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i + \frac{-f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}
 \end{aligned}$$

Hence, we get the formula for the Secant method,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Rubric:

- 4 points for a correct geometric derivation.
- 2 point for the correct formula.

- c) A *monotone function* is a function whose derivative never changes sign, i.e., the function is either always increasing or always decreasing. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a smooth monotone function with a root $x^* = k$, will the Newton–Raphson method always converge to this root, for any choice of the initial guess x_0 ? Justify your answer with a valid proof or a counter-example.

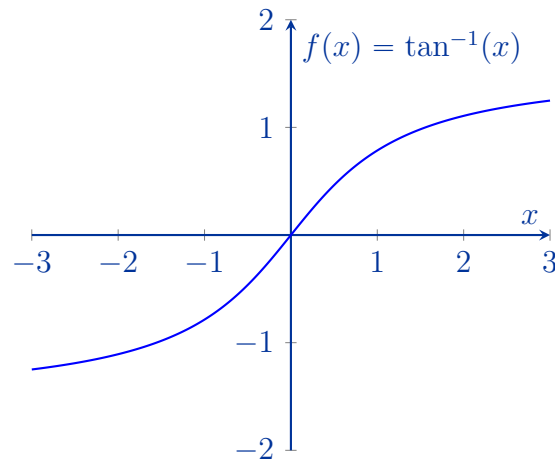
5
(CO3)
(PO1)

Solution:

The Newton–Raphson method will **not** always converge to the root x^* for any initial guess $x_0 \in \mathbb{R}$. A good counter-example to support this claim can be the function

$$f(x) = \tan^{-1}(x)$$

which can be graphically depicted as follows,



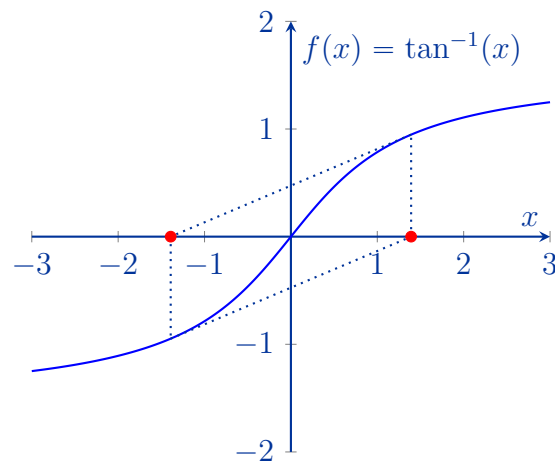
Now, if we start off with an initial guess x_0 that is close to the inflection point (which is also the root in this case), then we will see that the Newton–Raphson method will start to diverge away from the root after each iteration.

Rubric:

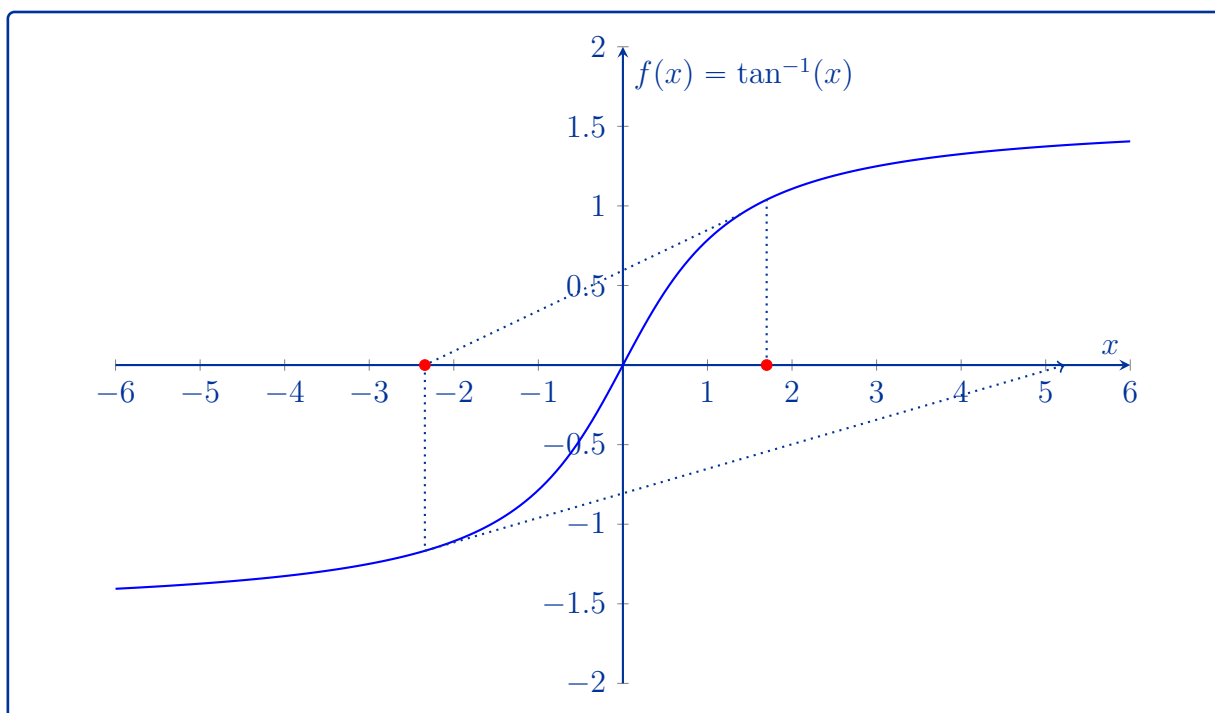
- 1 point for the correct verdict.
- 4 points for a valid counter-example function $f(x)$ and a valid initial guess x_0 .

Elaborate explanation:

It is obvious that a monotone function has a single root. Now, if that function has an inflection point or a platonic portion (also known as a saddle point) where $f'(x) = 0$, then the Newton–Raphson method runs into trouble. More specifically for the aforementioned counter-example function $f(x) = \tan^{-1}(x)$, it can be mathematically shown that if the starting guess x_0 is such that $|x_0| = 1.3917\dots$, then the algorithm gets stuck in an infinite loop and never converges to the root $x^* = 0$.



And, even worse, if the initial guess x_0 is such that $|x_0| > 1.3917\dots$, then the Newton–Raphson method gradually diverges from the root without any recourse.



- d) In the context of loan amortization, consider a scenario where an individual secures a loan of L dollars at an annual interest rate of r (in decimal form *i.e.* 5% is represented as 0.05). The loan is to be repaid over a duration of t years. Under these circumstances, the monthly payment amount, denoted as M , can be determined through the application of the *annuity equation*, which is mathematically expressed as follows:

$$L = \frac{12M}{r} \left[1 - \left(1 + \frac{r}{12} \right)^{-12t} \right] \quad (1)$$

- i. Formulate a function $f(r)$ such that its root will yield the value of the interest rate r for given values of L , M , and t .

2
(CO1)
(PO2)

Solution:

The given equation is,

$$L = \frac{12M}{r} \left[1 - \left(1 + \frac{r}{12} \right)^{-12t} \right]$$

For the given scenario, we need to express the equation as per the template $f(r) = 0$, and so the equation becomes,

$$L - \frac{12M}{r} \left[1 - \left(1 + \frac{r}{12} \right)^{-12t} \right] = 0$$

So the function $f(r)$ whose root yields the value of the interest rate r for given values of L , M , and t is,

$$f(r) = L - \frac{12M}{r} \left[1 - \left(1 + \frac{r}{12} \right)^{-12t} \right]$$

Rubric:

- 2 points for the correct function $f(r)$.

- ii. Suppose, you need to borrow \$150,000 to construct a new building, and you can only afford to pay \$600 per month. Assuming a 30-year mortgage, apply the Bisection method to estimate the interest rate (r) that you can afford to pay. The initial guesses are $r_l = 0.01$ and $r_u = 0.05$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations.

Table 1: The relevant values obtained in the answer of Question 1.d.

Iteration	r_l	r_u	r_m	$ \epsilon_a \%$	m	$f(r_m)$
1	0.01	0.05		—	—	
2						
3						
4						

Solution:

Here, $L = 150000$, $M = 600$, and $t = 30$. After performing the series of calculations, we end up with the following table.

Iteration	r_l	r_u	r_m	$ \epsilon_a \%$	m	$f(r_m)$
1	0.01	0.05	0.03	—	—	7686.371
2	0.01	0.03	0.02	50%	0	−12329.109
3	0.02	0.03	0.025	20%	0	−1852.256
4	0.025	0.03	0.0275	9.09%	0	3028.056

Rubric:

- 0.5 point for each approximately correct answer.
- 1 point for demonstrating the step-by-step procedure in each iteration.

2. a) Define Truncation Error. What are the different avenues through which Truncation Error can be introduced? Explain with proper examples.

5
(CO3)
(PO1)

Solution:

Truncation Error is the error caused by approximating the mathematical process or the modeling technique. Some of the ways it can be introduced are —

1. **Chopping off terms in a series:** If we consider only the first 3 terms of the Maclaurin series of e^x , we get $e^x \approx \left(1 + x + \frac{x^2}{2!}\right)$. And so the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= e^x - \left(1 + x + \frac{x^2}{2!}\right)\end{aligned}$$

2. **Using a finite step size instead of an infinitesimal value:** To find the derivative of a function $f(x)$ using a finite step size h instead of an infinitesimally small h , the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= \left(\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}\right) - \left(\frac{f(x+h) - f(x)}{h}\right)\end{aligned}$$

3. **Approximating area under curve with rectangles:** To find the integral value of a function $f(x)$ within given bounds a and b using n rectangles of width Δx_i and height $f(x_i)$, the truncation error is,

$$\begin{aligned}\text{Truncation Error} &= \text{Exact Value} - \text{Approximate Value} \\ &= \int_a^b f(x)dx - \sum_{i=0}^n f(x_i) \cdot \Delta x_i\end{aligned}$$

Rubric:

- 1 point for a correct definition.
- 2 points for each valid example. Any 2 valid examples will suffice.

- b) What is a transcendental function?

1
(CO2)
(PO1)

Solution:

Transcendental functions are functions that cannot be expressed using a finite sequence of terms or algebraic operations.

Rubric:

- 1 point for a correct definition.

- c) Euler's identity, which is $e^{i\pi} + 1 = 0$, is sometimes referred to as the most beautiful equation in mathematics. It is a special case of Euler's formula

$$e^{ix} = \cos(x) + i \sin(x) \quad (2)$$

17
(CO2)
(PO1)

when evaluated for $x = \pi$. One of the many proofs of this formula involves utilizing the Taylor expansions of its constituent transcendental functions.

The Taylor series for a function $f(x)$ is —

$$f(x + h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of —

- i. e^x ii. $\cos(x)$ iii. $\sin(x)$

Then use these three series to prove Euler's formula as delineated in Equation 2.

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(e^x) = e^x$, $\frac{d}{dx}(\sin(x)) = \cos(x)$, and $\frac{d}{dx}(\cos(x)) = -\sin(x)$.

- (i) Now, for $f(x) = e^x$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = e^x, f(0) = e^0 = 1; f'(x) = e^x, f'(0) = e^0 = 1; f''(x) = e^x, f''(0) = e^0 = 1; f'''(x) = e^x, f'''(0) = e^0 = 1; f^{(4)}(x) = e^x, f^{(4)}(0) = e^0 = 1;$$

Using the first 5 terms of the Taylor series,

$$f(x + h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$f(0 + h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow f(h) = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \frac{h^4}{4!} + \dots$$

$$\Rightarrow f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

- (ii) Now, for $f(x) = \cos(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \cos(x), f(0) = \cos(0) = 1; f'(x) = -\sin(x), f'(0) = -\sin(0) = 0; f''(x) = -\cos(x), f''(0) = -\cos(0) = -1; f'''(x) = \sin(x), f'''(0) = \sin(0) = 0; f^{(4)}(x) = \cos(x), f^{(4)}(0) = \cos(0) = 1;$$

Using the first 5 terms of the Taylor series,

$$f(x + h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 1 + 0 - \frac{h^2}{2!} + 0 + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(h) &= 1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots \\
 \Rightarrow f(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \\
 \Rightarrow \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots
 \end{aligned}$$

(iii) Now, for $f(x) = \sin(x)$, let's calculate the derivatives at the point $x = 0$.

$f(x) = \sin(x)$, $f(0) = \sin(0) = 0$; $f'(x) = \cos(x)$, $f'(0) = \cos(0) = 1$; $f''(x) = -\sin(x)$, $f''(0) = -\sin(0) = 0$; $f'''(x) = -\cos(x)$, $f'''(0) = -\cos(0) = -1$; $f^{(4)}(x) = \sin(x)$, $f^{(4)}(0) = \sin(0) = 0$; Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 0 + h + 0 - \frac{h^3}{3!} + 0 + \dots \\
 \Rightarrow f(h) &= h - \frac{h^3}{3!} + \dots \\
 \Rightarrow f(x) &= x - \frac{x^3}{3!} + \dots \\
 \Rightarrow \sin(x) &= x - \frac{x^3}{3!} + \dots
 \end{aligned}$$

Now,

$$\text{L.H.S} = e^{ix}$$

$$\begin{aligned}
 &= 1 + ix - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \dots \quad [\text{Since, } i^2 = -1, i^3 = -i, \text{ and } i^4 = 1.] \\
 &= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) + i\left(x - \frac{x^3}{3!} + \dots\right) \quad [\text{Separating the real and complex terms.}] \\
 &= \cos(x) + i\sin(x) \\
 &= \text{R.H.S.} \quad \square \text{ Q.E.D.}
 \end{aligned}$$

Rubric:

- 4.5 points for the correct series of e^x .
- 4.5 points for the correct series of $\cos(x)$.
- 4.5 points for the correct series of $\sin(x)$.
- 3.5 points for the correct proof of Euler's formula.

3. a) Compare and contrast the following interpolation methods: Direct method, Newton's Divided Difference method, and Lagrangian method.

4
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(A + b))$, or $O((n + 1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 2 points for each valid comparison.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, to approximate $f(x)$, you decide to use Newton's Divided Difference method of interpolation and obtain a quadratic interpolant $f_2(x)$. Derive the representation of 2nd order Newton's Divided Difference polynomial $f_2(x)$.

10
(CO4)
(PO1)

Solution:



Just [click here](#).

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 2 point for correct derivation of b_0 .
- 2 point for correct derivation of b_1 .
- 4 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in Table 2.

Table 2: Specific heat C of water as a function of temperature T for Question 3.c

Temperature, T ($^{\circ}\text{C}$)	Specific Heat, C ($\text{Jkg}^{-1}\text{C}^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Determine the value of the specific heat at $T = 61^{\circ}\text{C}$ using the Direct method of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be —

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}\text{C}$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations —

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have —

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{67} & \frac{91}{50} \\ -\frac{200}{67} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 \text{ Jkg}^{-1}\text{C}^{-1}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 2 points for the correct coefficient values.
- 2 points for the correct answer.

- d) Why is $f_9(x)$ in Figure 2(b) demonstrably worse at capturing the pattern of $f(x)$ compared to $S(x)$ in Figure 2(c)? Explain with necessary illustration(s).

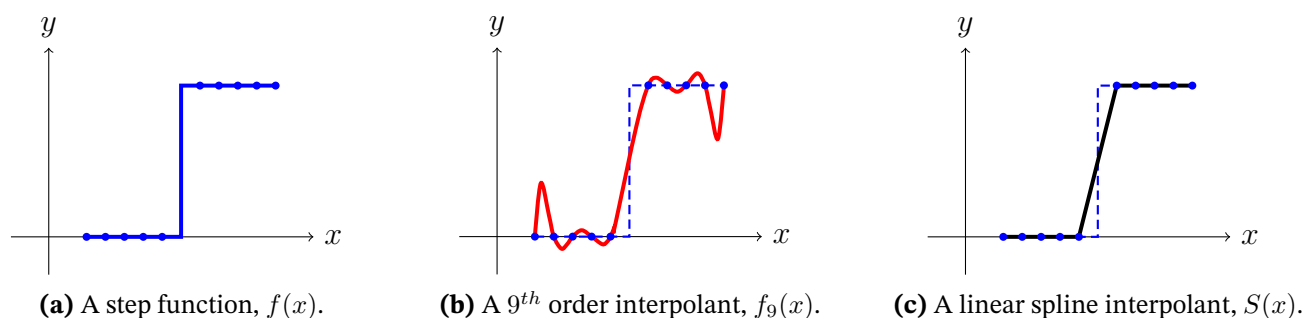


Figure 2: Predicting the nature of $f(x)$ for Question 3.d

Solution:

$f_9(x)$ is a worse interpolant than $S(x)$ due to the manifestation of the Runge phenomenon. Let's consider Runge's function $f(x) = \frac{1}{1 + 25x^2}$ to explain the phenomenon. When we try to approximate the nature of this function using a high-order interpolant, we observe an unhinged oscillatory behavior at either end of the interpolant.

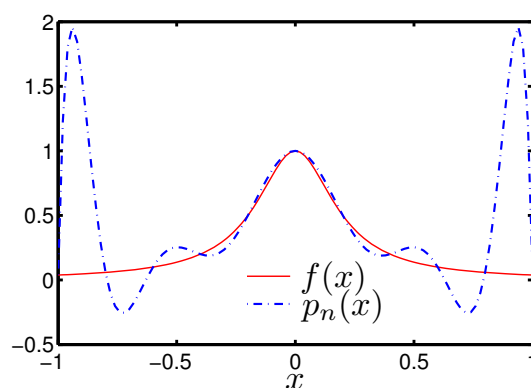


Figure 3: A 9th order polynomial interpolant passing through 10 equidistant points on Runge's function (**bad**).

However, we can ameliorate this phenomenon to a great extent by using splines.

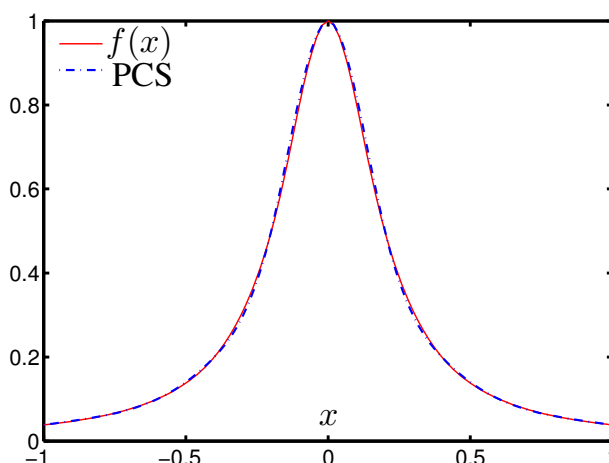


Figure 4: A 9th order cubic spline interpolant passing through 10 equidistant points on Runge's function (**good**).

As evident from the stark contrast between the interpolants portrayed in Figure 3 and Figure 4, splines are less susceptible to the Runge phenomenon since they stay true to the nature of the underlying function between their respective start and end points.

Rubric:

- 1 point for mentioning the correct reason.
- 4 points for illustrating nice figure(s) that explain(s) the Runge phenomenon.